

RQ5: Ablation to Bin size and GMM

Table 1 reports the mean IPR, F1, and AUC for every configuration evaluated in RQ5. Each value is the mean computed over six subject projects. Columns correspond to $k \in \{1, 2, 3\}$; rows correspond to method–Q pairs. Higher values are better for all three metrics. Bold entries indicate the best result within each metric column.

Figure 1. RQ5 parameter sensitivity comparison table, aggregated across all six projects. Each cell is the mean value over six projects. Columns correspond to $k \in \{1, 2, 3\}$; rows correspond to method–Q pairs. Higher IPR, F1, and AUC are better. [See rq5_comparison_table figure.]

Table 1. Parameter Configurations Comparison Table across 30 different settings ($k \in \{1, 2, 3\}$, $Q \in \{5, 10, 20, 50\}$), aggregated over six projects.

Configuration	IPR			F1			AUC		
	k=1	k=2	k=3	k=1	k=2	k=3	k=1	k=2	k=3
GMM	0.836	0.830	0.823	0.489	0.425	0.497	0.786	0.771	0.791
GMM+	0.912	0.907	0.909	0.513	0.513	0.514	0.799	0.798	0.799
BSA (Q=5)	0.831	0.822	0.817	0.498	0.508	0.512	0.795	0.801	0.806
BSA (Q=10)	0.828	0.819	0.816	0.501	0.512	0.514	0.796	0.802	0.808
BSA (Q=20)	0.827	0.823	0.812	0.506	0.511	0.516	0.795	0.801	0.806
BSA (Q=50)	0.831	0.821	0.811	0.508	0.507	0.514	0.795	0.798	0.803
BSA+ (Q=5)	0.876	0.873	0.871	0.516	0.516	0.516	0.803	0.803	0.804
BSA+ (Q=10)	0.870	0.860	0.860	0.518	0.519	0.520	0.804	0.804	0.805
BSA+ (Q=20)	0.864	0.857	0.850	0.518	0.520	0.519	0.805	0.805	0.805
BSA+ (Q=50)	0.856	0.841	0.840	0.519	0.519	0.518	0.806	0.806	0.806

Notes: Bold values indicate the best result within each metric $\times k$ column. Bold row labels (**BSA (Q=20)** and **BSA+ (Q=20)**) denote the recommended default configurations. GMM and GMM+ do not use the Q parameter.